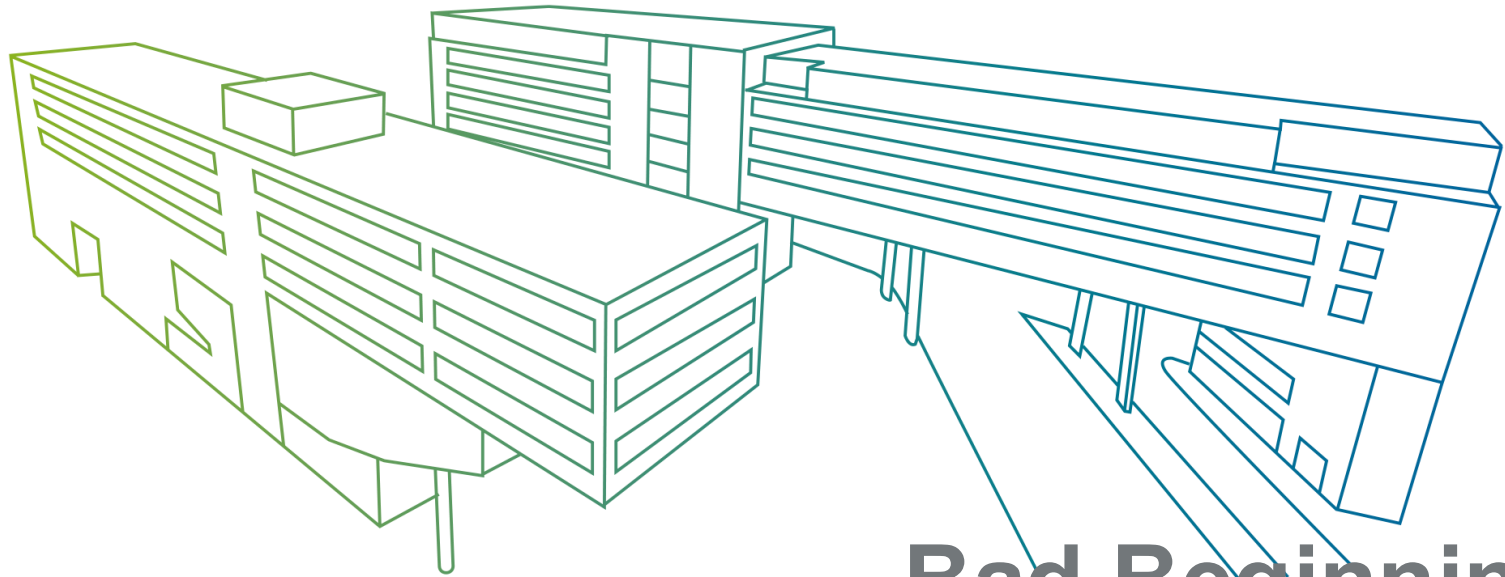




Bad Beginnings

Why the observation model, not the filter, dominates localization recovery under wrong initialization

FH University of Applied Sciences

TECHNIKUM

WIEN

Elias Bitsch · Probabilistic Robotics Lab, FH Technikum Wien

Task 2510331021 · Wrong Initialization

A wrong start breaks the filter

- Bayesian filters assume the start is roughly right.
- Reality breaks that assumption:
 - unknown power-on pose
 - a wrong operator estimate
 - a shove, the **kidnapped robot**
- Symmetric warehouses → **perceptual aliasing**: every aisle looks the same.
- The efficiency trick, collapsing to one guess, turns against you when that guess is wrong.



Does a fancier filter help, or does the measurement decide?

Two answers to “where am I?”

- **Scan-based** - a particle filter that weights poses by how well the laser scan matches the map (Nav2 AMCL)
- **Landmarks**, read identifiable references (AprilTags)

My contribution

- 5 filters × 7 wrong-init scenarios × 10 seeds
- 4 filters self-implemented in C++

I isolate the observation model: same motion, same wrong start, only the measurement changes.

Five filters, two ways to sense

Filter	What it measures	Belief	Source	Zustand	Geschwindigkeiten im Zustand?
KF	AprilTag landmarks	Single Gaussian	C++ (self)	6D: $(x, y, \theta, \dot{x}, \dot{y}, \dot{\theta})$	ja
EKF	AprilTag landmarks	Single Gaussian	C++ (self)	3D: (x, y, θ)	nein
EKF-LF	Lidar likelihood field	Single Gaussian	C++ (self)	3D: (x, y, θ)	nein
PF	Lidar likelihood field	Particle set	C++ (self)	3D: $(x, y, \theta) \times N$ Partikel	nein
AMCL	Lidar scan vs. map	Particle set	Nav2 baseline	3D: $(x, y, \theta) \times N$ Partikel	nein

Two read tags · three match the laser scan | *Gaussian = one hypothesis · particles = many*

APRIL Tags

A visual fiducial system

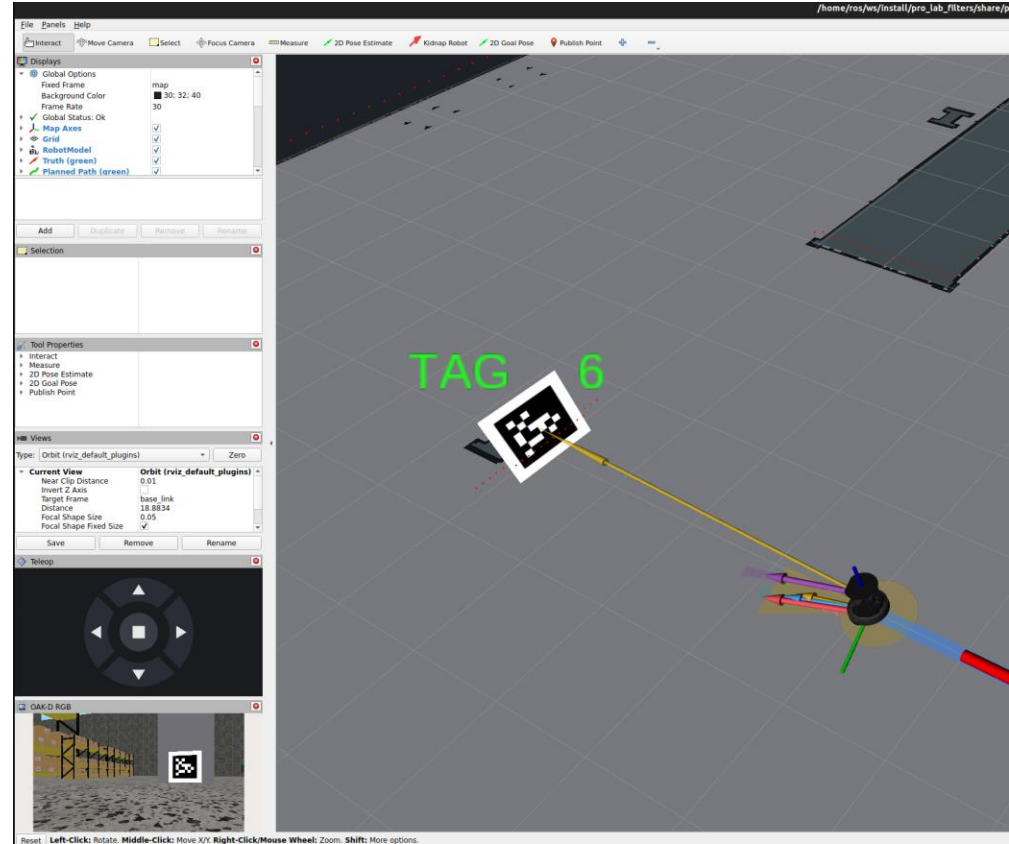
- Square 2D tags encoding a unique ID, like a simplified QR code
- Recover full 6-DOF pose from a single camera image
- Robust to lighting changes, rotation, and partial occlusion
- Standard tag families: 16h5, 25h9, 36h11
- Applications: robot localization, camera calibration, AR, autonomous landing



*APRIL Lab: Autonomy, Perception, Robotics,
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Method: the robot really reads the tags

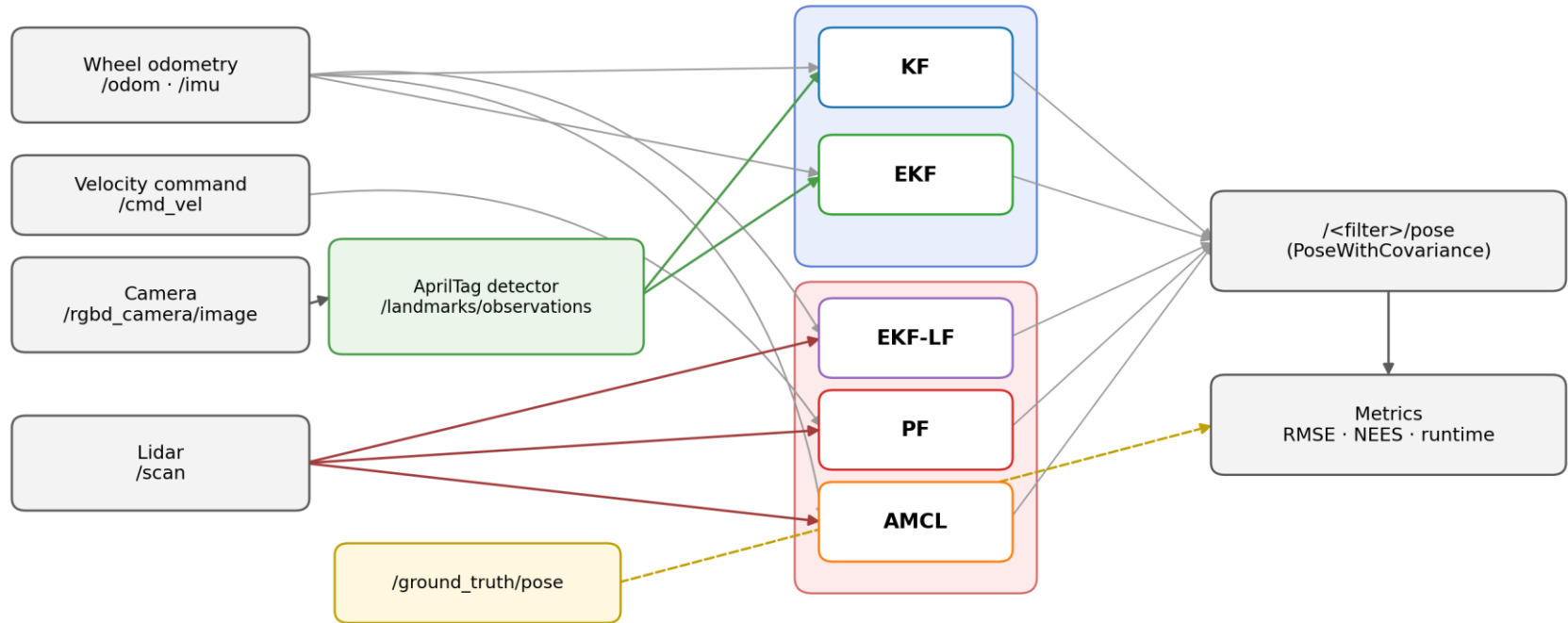
- TurtleBot 4 · ROS 2 Jazzy · Gazebo
- Symmetric 2×4 pillar grid, 7.5 m spacing
- Each pillar wears a unique **AprilTag 36h11**
- OAK-D camera reads ID + range + bearing
 - $\leq 1 \text{ cm}$ / $\leq 0.25^\circ$ vs. ground truth
- Identity is **measured**
- Ground truth only scores error, never a filter input



Reset · Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click/Mouse Wheel: Zoom. Shift: More options.

Dataflow

System data flow — ROS 2 topics into and out of each filter



Two read tags · three match the laser scan

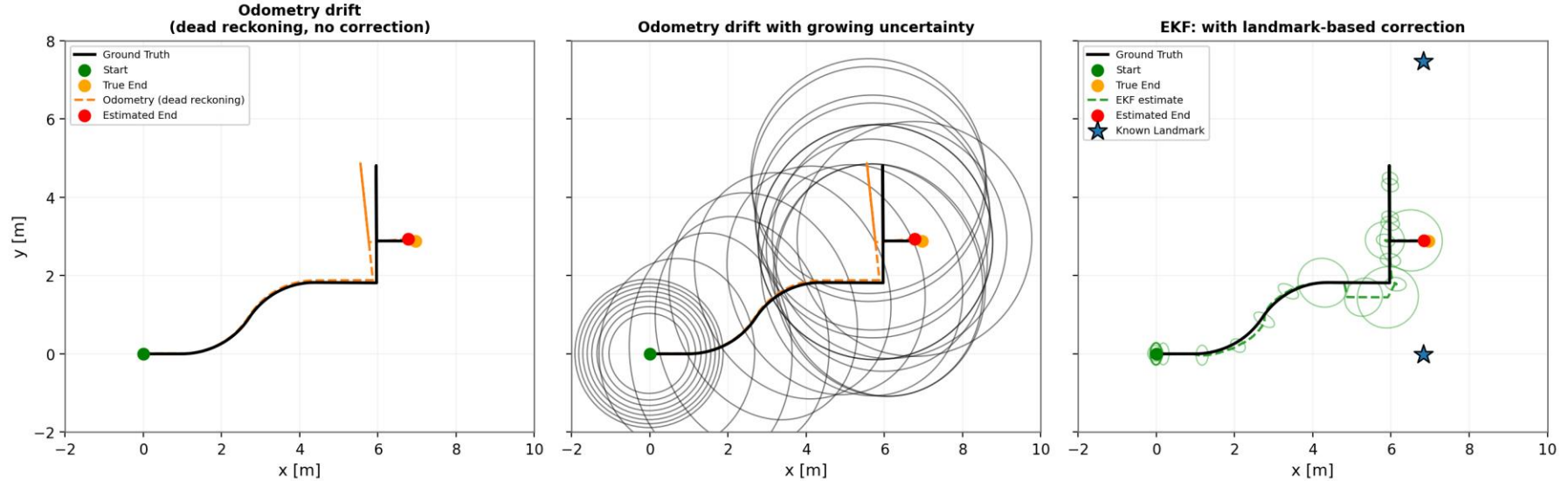
Gaussian = one hypothesis · particles = many

An experiment: same motion, only the sensor changes

- Identical motion, input topics, frames and scripted trajectory, **only the measurement modality differs**
- 7 wrong-init scenarios × 10 random seeds → report mean $\pm 1\sigma$
 - correct_init · offset_1m · offset_5m · wrong_yaw_pi2
 - overconfident_wrong · underconfident · kidnapped
- Metrics: final RMSE_xy · convergence rate · per-tick runtime + Big-O · NEES
- **Ground truth is used only to score error, never as a filter input**

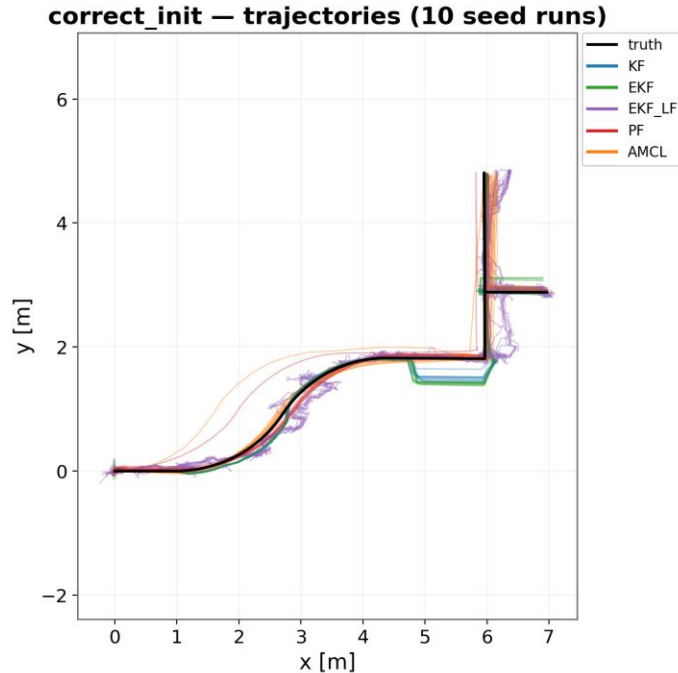
Why landmarks matter: bounded uncertainty

correct_init — EKF localization: odometry drift → growing uncertainty → landmark-based correction



Dead reckoning drifts, its uncertainty grows without bound. Landmarks collapse it and re-anchor the belief.

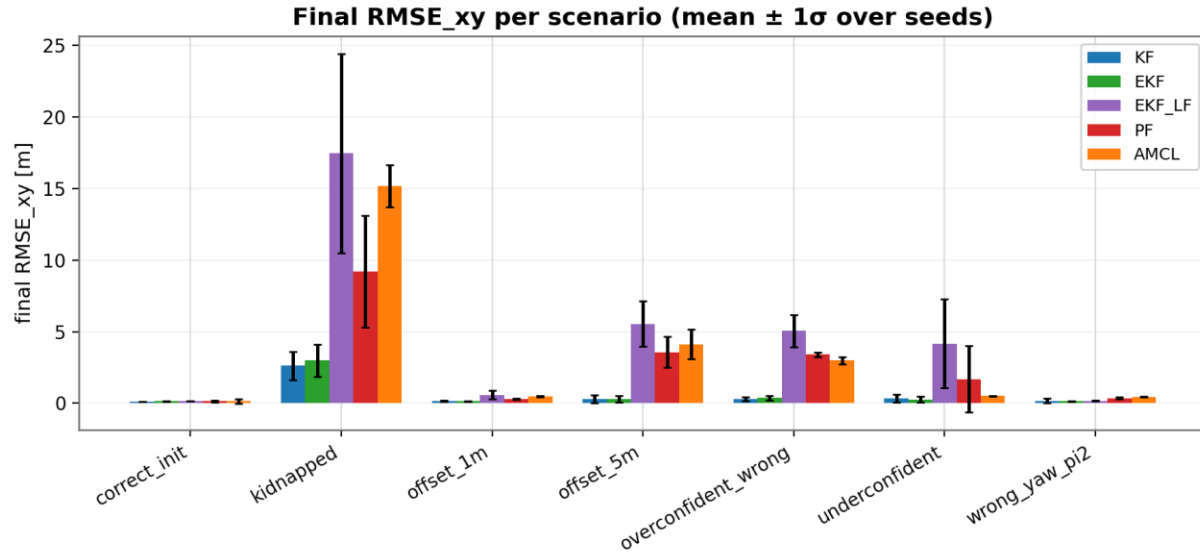
Correct start: everyone's equal



- All five filters track ground truth.
- Final error within **~15 cm** for every method.
- **The observation model doesn't matter yet.**

10 seed runs overlaid.

Recovery from a wrong start



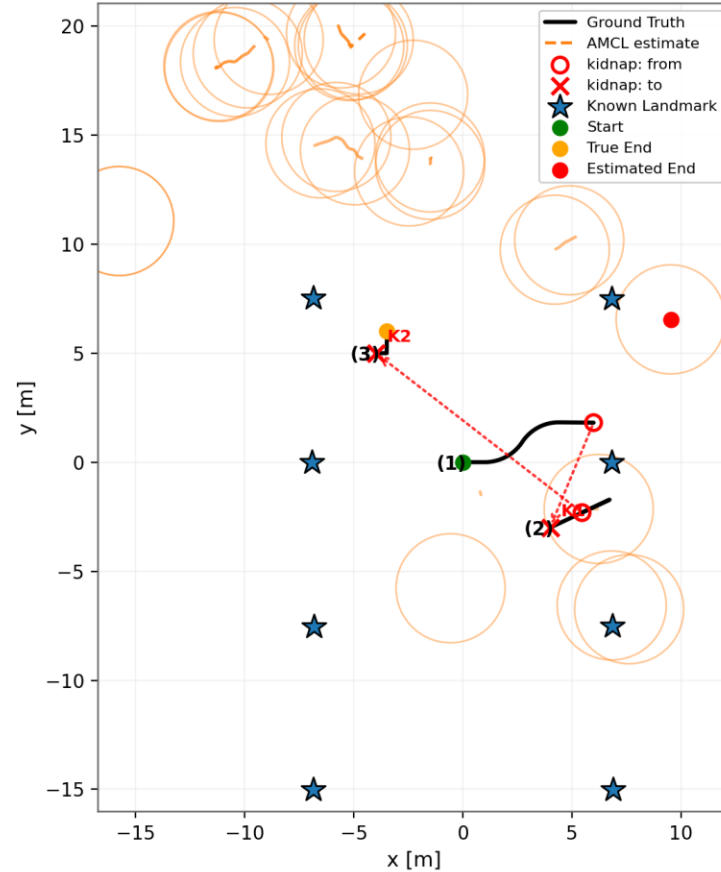
- CORRECT START -> all within 15 cm
- 5 m offset -> 0.3 m landmarks
-> 3 - 5 m scan

Wrong start → the modalities split hard.

The kidnapped robot

AMCL lost · up to 17 m

kidnapped — AMCL: trajectory + uncertainty (2 kidnap events)

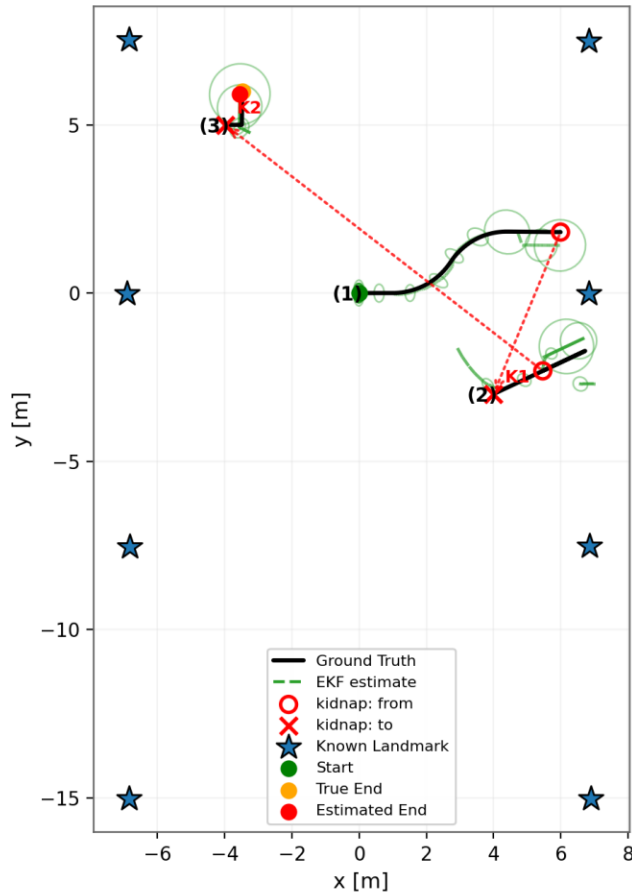


The kidnapped robot

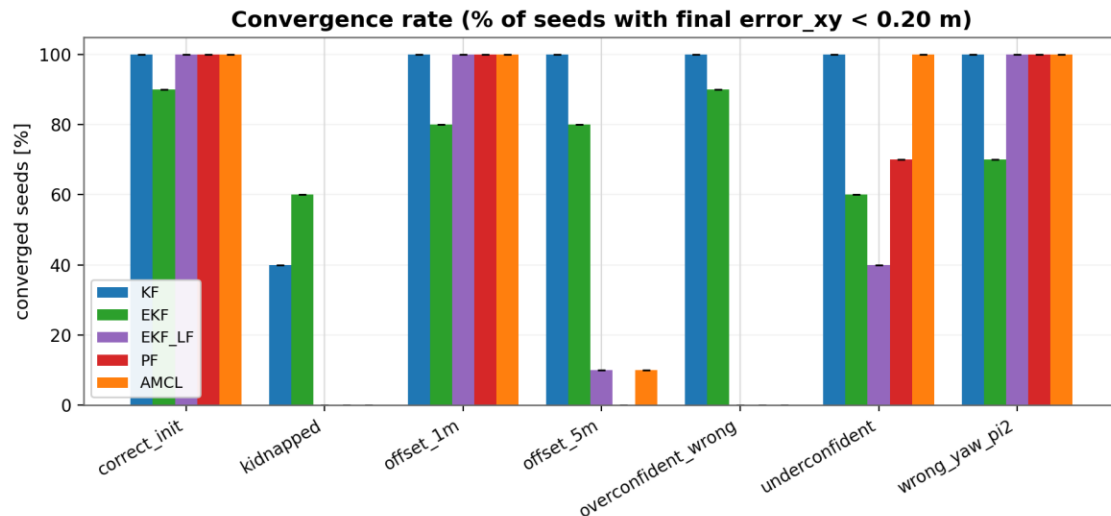
EKF recovers · ~2.6 m

Teleported twice, the EKF snaps onto the truth the instant it reads a tag; to AMCL every column looks the same.

kidnapped — EKF: trajectory + uncertainty (2 kidnap events)



Landmarks are king



- Perceptual aliasing → scan filters lock onto a wrong but symmetric hypothesis.
- **KF / EKF + Landmark recover autonomously.**
- PF / AMCL need a human “2D Pose Estimate”.

Under kidnapping: **KF/EKF 40–60% of seeds**; scan filters **0%**.

Even the textbook fix (augmented MCL / particle injection) fails here: the symmetric twin keeps scan-likelihood high, so the kidnap is never detected; injected particles just reconverge to a look-alike aisle.

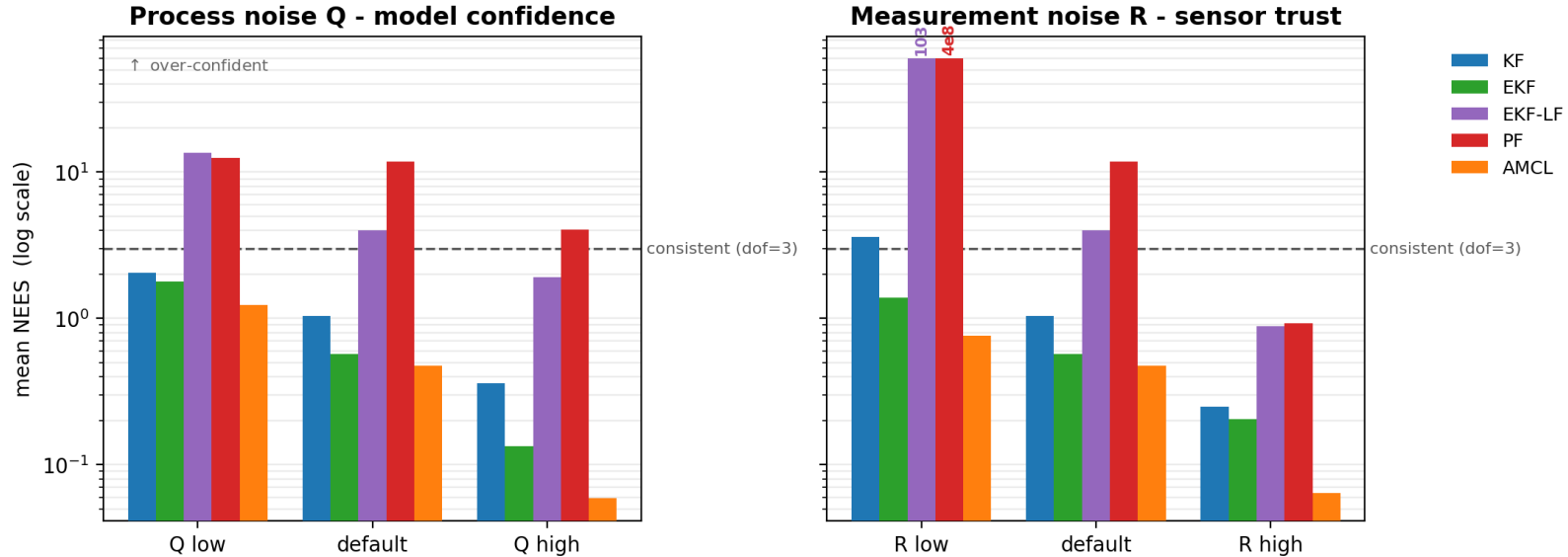
The textbook fix, and why it isn't enough

- We implemented the canonical kidnapped-robot remedy: **Augmented MCL** (Thrun §8.3) track short- vs long-term scan likelihood (w_{fast} / w_{slow}); when the likelihood drops, **inject random particles** to re-localize.
- But in our **symmetric** warehouse it still fails, for two reasons:
 - **Not even detected** - the look-alike aisle keeps the likelihood high, so no drop, no injection.
 - **Wrong convergence** - injected particles reconverge to a symmetric twin one bay over.
- → A *correct textbook implementation* is **not enough** under perceptual aliasing. You need **unique information** (fiducials) or **active perception**.



Noise tuning: motion model (Q) vs. sensors (R)

Filter consistency vs. process/measurement noise (10 seeds, NEES over x, y, yaw)



- Same correct start – we change only the noise, so it acts on confidence, not accuracy.
- Too little noise → over-confident (the PF even collapses); too much → sluggish. KF and EKF stay best balanced.

Live demo

Shown live in RViz

- 1 Kidnap the robot with the RViz 2D arrow
- 2 KF / EKF snap back on their own
- 3 Rescue PF / AMCL with the 2D Pose Estimate

Live demo

```
stlasi :~/git/PRO-LAB/scripts$
```

The observation model, not the filter, decides recovery

Don't reach for a fancier filter to survive a bad start, reach for an **identifiable measurement**.

- Landmark-reading filters bounce back from any bad start; scan-matching filters get lost in a symmetric warehouse.

Outlook

- Fuse laser scan + tags for the best of both
- Self-recovery in symmetric maps needs richer global cues than the likelihood field

